1. Introduction

In today's data-driven world, the demand for skilled professionals who can extract insights, generate models, or use data in some form is ever-growing. They look at tons of information and figure out how to use it to make businesses better. By studying data, they can predict what might happen in the future, make things work more efficiently, and even solve big problems like climate change and healthcare issues. So, these data jobs aren't just about making companies better, they are also about making the world a better place for everyone.

Understanding what affects how much money people in these data jobs make is super important. It's not just useful for people who want to be data experts, but also for companies that want to hire and keep the best talent. By looking closely at what influences the salaries, we can learn a lot about how pay works. This knowledge is also really important for how companies hire, keep their employees happy, and make sure everyone gets paid fairly in the world of data-driven jobs.

1.1. Aim

The main aim of this coursework is to conduct a comprehensive analysis of factors influencing salaries of data and AI related jobs using python programming language and its libraries.

1.2. Objectives

- To explore and understand the characteristics of the dataset.
- To preprocess and clean the dataset by removing unnecessary columns, handling missing values, checking for duplicates and so on.
- To perform statistical analysis on the dataset.
- To visually explore the dataset through different graphs and charts.

1.3. Tools and Libraries Used

a) Python

Python is an object-oriented, interpreted, and high-level programming language. It's easy and simple syntax makes it more readable and beginner friendly. Python comes with a large standard library that supports various tasks. It is most commonly used for developing websites, task automation, data analysis and visualization, machine learning, and artificial intelligence (Python Software Foundation, 2001-2023)

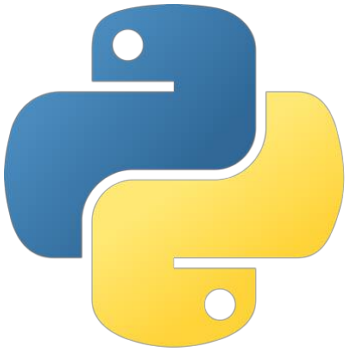


Figure 1: Logo of Python

Use:

- It is used as a programming language for the entire data analysis process.

b) Anaconda

Anaconda is a python distribution which is specifically for machine learning and data science. It comes with number of pre-installed packages like SciPy, Malplotlib, Pandas, and NumPy. It is also designed to work seamlessly with Jupyter Notebooks (Simplilearn Solutions, 2009-2024).



Figure 2: Logo of Anaconda

Use:

- It provides essential data science libraries and tools such as pandas, matplotlib without any manual installation.

c) Jupyter Notebook

The Jupyter Notebook is a web application for creating and sharing computational documents. You can write code in more than 40 programming languages, including Python.We can use big data tools like Apache Spark with Python, R, and Scala and also analyze data with tools like pandas, sci-kit learn and so on (Jupyter, 2024).



Figure 3: Logo of jupyter

Use:

- It is used as an interactive environment where Python code can be executed in a cell-by-cell manner for analysis process and its output.

d) Pandas

Pandas is a python library which mainly is used for managing datasets. It offers various functions for data analysis, cleaning, exploration, and manipulation. This library helps in the analysis of large datasets and the derivation of conclusions through statistical methodologies. With Pandas, messy datasets can be cleaned, making them meaningful and relevant (Refsnes Data, 1999-2024).



Figure 4: Logo of pandas

Use:

- It is used for loading the dataset into a DataFrame, handling missing values, removing unnecessary columns, and performing various data related querries.

e) Matplotlib

Matplotlib is a powerful libaray in python for making all sorts of graphs and charts. We can create high-quality plots for reports or presentations. It's easy to customize how our graphs look, and you can save them in different file formats. It also works well with other tools like JupyterLab, Jupyter Notebook and different graphical interfaces (Matplotlib, 2012-2023).



Figure 5: Logo of Matplotlib

Use:

- It is used for creating visualizations such as bar graphs, histograms, and box plots.

f) MS Word

Microsoft released MS Word, a program for making documents, on October 25, 1983. It helps create professional stuff like letters and reports. With its many features, we can easily edit and format files and documents (BYJU'S, 2021).



Figure 6: Logo of MS Word

Use:

- It is used for writing the report of the program.

2. Data Understanding

The dataset contains information about employees working in data-related jobs, including their experience level, job titles, salaries, and employment details like company size and remote work options. By understanding this data, we can see how salaries vary based on factors like experience level and job title. We can also explore how things like where the employee lives or the size of the company they work for affect their salary and whether they can work remotely. This helps businesses make decisions about hiring, compensation, and creating work environments that attract and retain talent in the data field.

The column description of each column present in the dataset is given in the table below:

S. N	Column Name	Description	Data Type
1	work_year	This column provides the year of the data-related job. It ranges from 2020 to 2023.	Integer
2	experience_level	This column contains values indicating the experience level i.e. “SE” for senior level, “MI” for medium level, “EN” for Entry Level and “EX” for executive level.	Object
3	employment_type	The 'employment_type' column contains values indicating the type of employment namely 'FT' for full-time, 'CT' for contract, 'FL' for freelance, and 'PT' for part-time.	Object
4	job_title	This column specifies the titles or designations associated with each employee.	Object
5	salary	The ‘salary’ column’ provides the salary(number) of the job. The salary might be in any currency.	Integer
6	salary_currency	This column contains the currency of salary in ‘salary’ column. For example, if the	Object